An autonomous career system for embodied creative research and development

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Autonomous agents are increasingly able to plan, monitor, and execute tool-mediated workflows, but a career objective is normally treated as motivation rather than as a measurable control problem. We report an autonomous career system that converts a target role, Walt Disney Imagineering Research and Development mechanical Imagineering in Glendale, into a logged state machine with role-fit dimensions, guardrails, review cycles, weekly manuscript updates, and public artifacts. The review loop uses GPT-5.5 and a whole-public-AO-Labs source graph, including the CV, Sarrus, FluxCell, Relay, Relay Live, Ocean, other public project surfaces, the WDI R&D profile, the active WDI listing, and Disney Research context. It can critique public sources autonomously but cannot send outreach, claim relationships, submit applications, or use private email/contact data without approval. At the current lock on 6 May 2026, the system tracked six role-fit dimensions, four portfolio anchors, seven logged work items, one daily cycle, ten AI reviews, zero outreach events, five public artifacts, and twenty journal entries. The public dashboard is report-first: the primary section is the AI-generated readout of Alan's current public record, followed by review-system status and the manuscript artifact. After recalibration, the live fit signal is 74 of 100 with a credible-but-needs-signal confidence label, while the autonomous review scores the public profile 88 of 100 after the system published a Sarrus mechanism page, added an HTML-visible measurement table, and inserted role-calibration boundaries for the active Glendale mechanical-design rung versus the Principal R&D Imagineer north-star profile. The current bottleneck is review intelligence and level calibration: the strongest public record supports active WDI mechanical-design fit, while principal-level leadership scope and an active principal posting remain unverified. The contribution is therefore not a claim that the system has secured a job. It is a falsifiable protocol for making an otherwise ambiguous career-conversion process auditable, adaptive, and constrained by truthful public sources.

1 Introduction

Tool-using language-model systems can now reason, call external services, write artifacts, and update deployed software (*1–4*). These capabilities create an opportunity to treat long-horizon personal objectives as monitored systems rather than as episodic intentions, in the older spirit of autonomic systems and closed-loop discovery workflows (*5*, *6*). A career target is especially difficult because the desired outcome depends on evidence quality, technical fit, social trust, timing, applications, and human judgment. Without a state model, repeated effort can feel intense while remaining scientifically uninterpretable.

This paper describes an early deployment of an autonomous career-signal system aimed at WDI Research and Development mechanical Imagineering roles. The system uses the active WDI Research and Development Imagineer - Mechanical Design Engineer role as the live rung and the Principal R&D Imagineer - Mechanical Engineer profile as the north-star target (*7*). The positioning line is deliberately stable: mechanical PhD, soft robotics, creative prototyping, and AI-assisted tools for physical interaction systems.

The design borrows from controlled revenue experimentation: define a measurable state, choose the current bottleneck, constrain allowed actions, record interventions, and update a paper from real source material rather than from aspiration. The system is not allowed to invent credentials, inflate relationships, send spam, or apply on behalf of the operator. It can monitor state, update dashboards, maintain a paper, identify the next constraint, run a source-bounded AI review, and create or improve public artifacts that are grounded in existing work.

2 System architecture

The deployment consists of a public dashboard, a Railway-hosted backend, a JSON runtime state file mounted on a persistent volume, and scheduled or operator-triggered automation. The backend exposes health, operations-check, journal, weekly-paper, event, daily-cycle, and AI-review endpoints. The dashboard is intentionally minimal and report-first. Its three sections are: an AI-generated readout on Alan's current public record, a compact status surface for the review system and role-fit dimensions, and the paper/PDF artifact. The dashboard does not treat the review loop as the main subject; it treats it as an instrument that generates feedback for human judgment.

Table 1. State variables at the current evidence lock. Values are reported from the live system after deployment of the whole-public-AO-Labs GPT-5.5 reviewer on 6 May 2026.

Variable	Recorded value
Target company and location	Walt Disney Imagineering R&D, Glendale, California
Live rung	WDI Research and Development Imagineer - Mechanical Design Engineer
North star	Principal R&D Imagineer - Mechanical Engineer
Fit signal	74 of 100 with credible-but-needs-signal confidence
Dominant bottleneck	Review intelligence: repeatable critique, public-source depth, and approved escalation
Counters	Seven logged work items; one daily cycle; ten AI reviews; zero outreach events; four portfolio anchors; five public artifacts; twenty journal entries
Active experiment	Autonomous AI reviewer v0
Reviewer source graph	Twenty-five live sources in the latest production run, spanning public AO Labs projects and WDI/Disney Research context
Human approval boundary	Applications, sensitive outreach, referrals, claims involving real people, and any private contact/email data

The reviewer is a separate loop rather than a human-contact substitute. It builds a source bundle from the live state, the target listing, the WDI R&D profile, the paper PDF, the CV, Sarrus, FluxCell, Relay, Relay Live, Ocean, Talk, Nerve, Duet, Violin, Yum, Lily, AO Labs home, Disney Research roster, and a Disney Research bipedal-character paper page. It also includes a compact identity profile summarizing the operator as a mechanical engineering PhD candidate building soft robotic materials, reconfigurable mechanisms, morphing interfaces, research papers, public project surfaces, and autonomous systems. The model returns a verdict, score, current constraint, strongest signals, unresolved signals, profile updates, and candidate actions as JSON. If the model fails or times out, a deterministic fallback reviewer produces a conservative critique and records the fallback reason.

3 Decision policy

The policy scores six role-fit dimensions: mechanical depth, creative prototyping, human-facing motion, review intelligence, Glendale application profile, and the paper/system itself. Each dimension now reports what it tracks, its basis, and the unresolved signal that would move the score. The score basis is intentionally calibrated: a base prior from the state file, a bounded event bonus, a bounded portfolio bonus, a daily-cycle bonus, optional review-loop credit, and a readiness ceiling that names the unresolved blocker. A score of 100 is reserved for no known blocker in that lane, not for accumulated activity. The candidate action is chosen from the weakest current signal. In the current lock, the weakest signal is review intelligence. The selected intervention is to improve public-source depth and prepare human-review material without sending outreach before approval.

This policy is intentionally conservative. It can make public artifacts, update the dashboard, refresh the manuscript, run autonomous critique, and record state. It cannot manufacture the social trust that a principal-level role requires. If an action requires a real person's attention or reputation, the system must stop at preparation and request approval before sending. The review loop therefore becomes a repeatable calibration instrument, not a license to automate human outreach.

4 Results

The current deployment is functional and has moved beyond the initial dashboard-only state. The dashboard and backend are live, the weekly paper endpoint can publish a snapshot, the site exposes a PDF manuscript, and the AI reviewer has produced live production critiques from the public source graph. The latest reviewer run used GPT-5.5, consumed 25 sources, returned a review score of 88 of 100, and identified a narrower dominant issue: the strongest public record now supports the active Glendale WDI Research and Development Imagineer - Mechanical Design Engineer rung, while the Principal R&D Imagineer north-star remains unverified.

The largest artifact change in this cycle was the Sarrus mechanism page. The system published a page with mechanism geometry, compliant-link and flexural-joint path, pressure-driven expansion, module assembly, build state, force/stiffness figures, and motion examples. It then added an HTML-visible measurement table with pressure range, expansion ratio, estimated leg angle, output force, local stiffness, hysteresis status, planar dome height, double-dome resolution, and prototype limitation. The WDI R&D profile now uses this table as the dominant technical path while preserving the current data boundary: raw digitized force, stiffness, and hysteresis curves with uncertainty remain unresolved.

After the latest interface revision, the live site no longer leads with generic status tiles. It leads with the AI readout: verdict, review score, summary, current constraint, strongest signals, unresolved signals, profile updates, and the selected candidate action. The review-system section remains visible for operational transparency: model, source count, review count, approval boundary, fit signal, bottleneck, counters, and role-fit score explanations. This distinction matters because the useful artifact for the

Table 2. Latest autonomous review result. The reviewer is a GPT-5.5 source-bounded critique over the public AO Labs graph, not a human recommendation.

Field	Recorded value
Model	GPT-5.5 via the Responses API
Scope	Whole public AO Labs graph plus WDI role and Disney Research context
Source count	25 sources in the latest production run
Review score	88 of 100
Verdict	Strong active-rung fit; principal scope not yet verified
Top issue	Level calibration: the strongest public record supports the active Glendale Mechanical Design Engineer rung, while the principal north-star remains unverified
Next action	Keep the role-verification banner above the fold

operator is the critique, not a status page about the critique generator.

The system-level fit signal is now calibrated to 74 of 100 after removing the earlier saturation behavior that let logged work push several lanes to 100. This should be interpreted as an internal readiness signal, not as an external hiring probability. The review result is deliberately separate: it scores the current public profile, not the probability of being hired. It confirms that the overall direction is credible for WDI R&D mechanical design while rejecting the idea that the current public profile is already principal-level. The profile now includes both a principal-scope boundary and an above-fold role-verification banner: the active target is the Glendale mechanical-design rung; the principal title remains a north-star profile until an active principal posting and principal-scope leadership work are independently verified.

5 Limitations and guardrails

The system cannot guarantee an offer. A job decision depends on timing, hiring needs, competition, interview performance, portfolio interpretation, and institutional context. The useful claim is narrower: the system can make progress legible, choose the next bottleneck, prevent fake progress, and turn repeated effort into logged artifacts. It should be evaluated by public artifact velocity, critique quality, application readiness, interviews, and eventual conversion.

The guardrails are part of the result. The system forbids fabricated credentials, fake outreach, invented Disney relationships, automated applications, and claims that are not supported by public or logged

Table 3. Role-fit dimensions after the current autonomous review cycle. Scores are internal operating metrics, not external hiring probabilities; the live dashboard exposes the tracked signal, basis, and unresolved signal for each row.

Dimension	Score	Current interpretation
Mechanical depth	86	Strong but not complete; Sarrus has a public mechanism page and measurement table, with raw digitized curves, uncertainty, and test conditions still pending.
Creative prototyping	82	Strongest signal; should remain attached to concrete physical demonstrations.
Human-facing motion	79	Credible signal, but still needs a concise guest-facing motion sequence.
Review intelligence	56	Current bottleneck; review loop works, but public-source depth, human-review material, and leadership scope remain weak.
Glendale profile	70	Credible for the active rung, but still needs demo reel, final CV bullets, and principal-scope work.
Autonomous career system	70	Backend, dashboard, durable state, reviewer, journal, and paper loop are live, but scheduled runs, evaluation history, and outcome tracking are early.

sources. Private email/contact data is excluded from the reviewer unless explicitly approved. These constraints reduce short-term aggressiveness, but they preserve interpretability. If the system ever reports strong progress, that progress should be traceable to artifacts, logs, AI reviews, human approvals, applications, or outcomes.

6 Availability

The live dashboard is available at `https://imagineer.aolabs.io/`, with a fallback at `https://aolabs.io/imagineer/`. The live backend exposes `https://imagineer.aolabs.io/api/imagineer/ops-check`, `https://imagineer.aolabs.io/api/imagineer/weekly-paper`, and `https://imagineer.aolabs.io/api/imagineer/ai-review`. This manuscript is a public artifact and should be refreshed from logged sources as the system accumulates real outcomes.

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